

Karunadu Technologies Private Limited

Internship Project



Project Title

Lung Cancer Prediction using Machine Learning & Django



Introduction

Lung cancer is one of the leading causes of death worldwide. Early prediction of lung cancer risk based on behavioral and health-related factors can help in early diagnosis and treatment.

In this assignment, students will develop a **Machine Learning-based web application** to predict whether a person is likely to have **lung cancer** based on survey data including lifestyle habits, symptoms, and medical conditions.

.



Dataset

The dataset contains various attributes such as:

1. Gender (Male/Female)
2. Age
3. Smoking (Yes/No)
4. Yellow Fingers (Yes/No)
5. Anxiety (Yes/No)
6. Peer Pressure (Yes/No)
7. Chronic Disease (Yes/No)
8. Fatigue (Yes/No)
9. Allergy (Yes/No)
10. Wheezing
11. Alcohol Consuming
12. Coughing
13. Shortness of Breath
14. Swallowing Difficulty
15. Chest Pain

Note: Interns will build and compare multiple classification models to predict the output label.



Objective

- Apply Machine Learning classification algorithms
- Understand data preprocessing and feature engineering
- Build a web application using Django
- Compare performance of different algorithms

Backend

Python
Django
Framework
Scikit-learn

Frontend

HTML
CSS / Bootstrap
Django Templates

Expected Output

- User login system
- Data input form
- Disease prediction output
- Accuracy comparison of models



Tasks

Interns must implement **at least 4 classification algorithms** from the list below:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes



Tasks

Web Application (Django)

Interns must build a **full-stack web application** using Django.

Login Module

- User Registration
- Login Authentication
- Session Management

Input Form Page

Build Django UI with input form of all Inputs

Prediction Module

Add a **“Predict” button**

On submission:

- Load trained ML models
- Predict output label
- Display result on UI

Results Display

- Predicted Output
- Accuracy of each algorithm
- Best performing algorithm



Important Questions

- What is classification in Machine Learning? How is it different from regression?
- Explain the importance of the output in this dataset.
- What is feature selection? Why is it important in this project?
- Explain the difference between supervised and unsupervised learning.
- What is overfitting and underfitting? How can it be avoided?
- What is the role of training and testing datasets?
- Explain how Logistic Regression works for classification.
- How does a Decision Tree decide splits?
- What is the advantage of Random Forest over Decision Tree?
- Explain how KNN works and its limitations.
- What is the concept behind SVM (Support Vector Machine)?
- What is a Confusion Matrix? Interpret it for your model.
- What preprocessing steps did you apply to the dataset?
- How did you handle missing or inconsistent data?
- Explain the flow of your Django application.
- How does authentication (login system) work in your project?
- What is the role of views, templates, and models in Django?
- How is the prediction triggered from the frontend?



Submission Requirements

Interns must submit

- Internship Report
- Source Code (Django + ML models)
- Dataset (if modified)
- Project Report (PDF) including:
 - Problem Statement
 - Methodology
 - Algorithms Used
 - Results & Comparison
 - Screenshots



Contact US



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**Thank You Interns for being a
part of Internship Journey**

**Contact Karunadu Technologies Private Limited
For Internship**